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INTERNSHIP REPORT

**IoT Network Traffic Generator with IDS/IPS & PCAP
Logging
using Python**

Submitted by

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Declaration

I, **Amjad Khan**, declare that this internship report on “**IoT Network Traffic Generator with IDS/IPS & PCAP Logging**” is my own work, done as part of the AICTE Smarthan Internship Connect 2026–2027 at IIITDM Kancheepuram, under the guidance of **Dr. Bhale Pradeepkumar Gajendra**.

All the ideas, code, testing, and observations written in this report are based on the work I personally carried out during the internship period. I have not submitted this report to any other institution or used it for any other academic purpose.

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I also want to thank the faculty and staff of the Department of Computer Science and Engineering, IIITDM Kancheepuram, for their support. Working here taught me things that go beyond what is written in textbooks, and I am genuinely grateful for that.

Amjad Khan

Abstract

IoT devices are now being used in almost every industry — from factory floors and hospital wards to smart city infrastructure. As the number of connected devices keeps growing, network security has become a much bigger problem. Testing whether a security system actually works requires sending realistic network traffic, including both normal data and attack patterns. Most available tools can only handle one or two protocols at a time, which is not enough for a proper IoT setup.

This report describes the design and development of a **Multi-Protocol IoT Network Traffic Generator with Active IDS/IPS and PCAP Logging**, built entirely in Python during my AICTE Smarthan Internship at IIITDM Kancheepuram. The tool can run hundreds of simulated IoT devices at once, each sending data through **HTTP, TCP, UDP, or MQTT**. The traffic mode can be switched between normal operation, burst traffic, and DDoS attack using a `config.json` file — without stopping the program.

A central receiver module (`receiver.py`) collects all incoming data, logs it into a Wireshark-compatible `capture.pcap` file via `pcap_logger.py`, and runs a built-in **IDS/IPS engine** that checks each packet for rate-based flooding and known attack strings like `OVERFLOW_ATTACK`. Any device that crosses the 12 packets/sec limit or sends a suspicious payload is immediately blocked. A live web dashboard (`dashboard.html`) shows charts, protocol statistics, and security alerts in real time. During my tests, the system handled over 1,200 packets per second with low CPU usage, which showed that it works reliably even under heavy load.

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List of Abbreviations

IoT	Internet of Things
IDS	Intrusion Detection System
IPS	Intrusion Prevention System
PCAP	Packet Capture
MQTT	Message Queuing Telemetry Transport
HTTP	Hypertext Transfer Protocol
TCP	Transmission Control Protocol
UDP	User Datagram Protocol
DDoS	Distributed Denial of Service
JSON	JavaScript Object Notation
API	Application Programming Interface
IP	Internet Protocol
AICTE	All India Council for Technical Education
IIITDM	Indian Institute of Information Technology Design and Manufacturing

1. Introduction

IoT devices have become a major part of how modern systems work. Factories use them for automation, hospitals use them to monitor patients, cities use them to manage traffic and power grids, and homes use them in everything from smart bulbs to security cameras. All of these devices connect to the internet and constantly exchange data. Managing and securing this kind of network is not simple, and three main challenges make it harder:

- 1. Heterogeneous Application Protocols:** IoT devices do not all use the same protocol. Gateways connected to web services usually send data using HTTP POST requests. Small battery-powered sensors prefer UDP because it is faster and lighter. Devices that need to stay connected over time use TCP streams. Remote sensors in low-bandwidth setups often use MQTT, which is very efficient. Having all of these running together in one network makes simulation much harder.
- 2. Unpredictable and Burst-Heavy Traffic Patterns:** IoT traffic is not steady. A smart meter may send one reading per hour, but a fire alarm system might send hundreds of alerts within a few seconds if something goes wrong. This kind of burst behavior is difficult to replicate using tools built for regular web traffic.
- 3. Escalating Cybersecurity Vulnerabilities:** A large number of IoT devices are shipped with weak default settings, old firmware, and no encryption. Attackers often take advantage of this to build botnets — the Mirai botnet is a well-known case where hundreds of thousands of IoT devices were used to launch DDoS attacks on major internet services. Defending against such threats requires tools that can actually simulate them in a controlled way.

Building and testing real defenses against these problems requires a proper testbed. Buying hundreds of physical IoT devices just for security testing is not practical for most research teams. A software-based traffic simulator that can mimic these conditions on a single machine is a much more useful option.

1.1. Project Scope and Motivation

The goal of this project was to build a **multi-protocol IoT Network Traffic Generator** using Python, with no external library requirements, so that it can run on any standard computer. The complete system consists of:

- A traffic generator (`generator.py`) that creates multiple simulated IoT devices, each sending data through HTTP, TCP, UDP, or MQTT.

- A central receiver (**receiver.py**) that handles all incoming connections at the same time and checks each packet for security threats.
- A packet capture module (**pcap_logger.py**) that saves raw network frames into a **.pcap** file that can be loaded in Wireshark.
- A web-based dashboard (**dashboard.html**) that shows live traffic charts and security alerts in the browser.

2. Internship Objectives

The primary milestones established for this AICTE Smarthan Internship project under the guidance of Dr. Bhale Pradeepkumar Gajendra at IIITDM Kancheepuram focused on the following targets:

1. **Multi-Protocol Simulation Engine:** Research and construct concurrent client-side traffic generator pipelines via native Python sockets and networking utilities to handle four cornerstone IoT protocols: HTTP, TCP, UDP, and MQTT.
2. **Dynamic Configuration and State Management:** Build a centralized JSON layout scheme (`config.json`) enabling runtime updates to endpoint counts, target addresses/ports, packet transmission intervals, payload templates, and simulation states without terminating the process.
3. **Traffic Profile Emulation:** Formulate and deploy three unique behavioral profiles:
 - **normal:** Constant, low-frequency sensor telemetry reporting.
 - **burst:** Rapid telemetry transmissions pushing spiked values that represent environmental irregularities.
 - **attack:** High-speed DDoS packet streams embedded with synthetic botnet threat signatures.
4. **Centralized Multi-Protocol Receiver & Telemetry Aggregator:** Architect a multi-threaded backend server configured to intercept data on distinct ports for all four core protocols simultaneously, log throughput values in real time, and map RESTful endpoints (`/api/stats`).
5. **Active Intrusion Detection & Prevention System (IDS/IPS):** Build a practical, real-time security management system directly within the receiver application to:
 - Monitor device-specific packet rates across a 5-second sliding window and flag any device sending more than 12 packets/sec as a DDoS source.
 - Scan incoming data for known attack strings such as `OVERFLOW_ATTACK`.
 - Automatically add flagged devices to a blocklist and cut off their connections at the socket layer (403 Forbidden for HTTP, direct termination for TCP/UDP/MQTT).
 - Provide a manual toggle (**Active Drop vs. IDS Only Mode**) on the dashboard to switch between blocking mode and monitoring-only mode.

- 6. Pure-Python PCAP Packet Capture Engine:** Write a low-level packet logging module (`pcap_logger.py`) that builds binary PCAP files from scratch using Python's `struct.pack`, with correct Ethernet, IPv4, TCP, and UDP headers and checksum values that load cleanly in Wireshark.
- 7. Interactive Web Analytics Dashboard:** Build a dark-themed HTML5/JS monitoring page (`dashboard.html`) at `http://127.0.0.1:5007/dashboard` where the user can view live traffic charts, read security alert logs, and change the simulation mode at any time.

3. Literature Review

Before starting to code, I looked into several existing tools that are used for network traffic generation and packet crafting. Understanding what these tools can and cannot do helped me decide what my own tool needed to handle differently.

3.1. `hping3`

`hping3` is a command-line packet tool that lets you manually build and send packets over TCP, UDP, ICMP, and RAW-IP. It works well for simple tasks like testing firewalls or checking if a port is open. However, it has no support for running automated, multi-device simulations. You cannot use it to run 50 simulated IoT sensors at the same time, and it does not support higher-level protocols like HTTP or MQTT. So while it is useful for low-level testing, it is not suitable for IoT traffic simulation.

3.2. `Ostinato`

`Ostinato` is an open-source packet generator with a graphical interface and Python API support. It gives users detailed control over packet structure and works well for hardware-level throughput testing. The main problem is that it only sends stateless packets — it does not maintain open connections. For IoT simulation, you need tools that can keep a TCP connection alive or hold an MQTT session open, and `Ostinato` cannot do that.

3.3. Open Traffic Generator (OTG)

OTG is a standard API designed for automating network tests in large enterprise and cloud setups. It is well-suited for data center environments where traffic needs to be tested at scale. Setting it up, however, requires a lot of configuration and additional software components. For a single-machine project like this internship, the setup overhead makes it impractical.

3.4. `TRex`

`TRex`, developed by Cisco, is a high-performance traffic generator that uses DPDK to send traffic at multi-gigabit speeds. It supports complex stateful flows from layer 4 to layer 7. The limitation is that it requires specific network interface cards and Linux kernel-level configuration. It simply cannot be used on a regular Windows laptop or office workstation, which made it unsuitable for this project.

3.5. Scapy Python Framework

Scapy is a Python library that can craft and send packets for almost any network protocol. I looked into using it, but found that it slows down considerably when sending a large volume of packets. In my own testing, it could not keep up with the packet rates needed for simulating a DDoS attack. This made it too slow for the attack simulation part of the project.

3.6. GothX

GothX and tools like it are designed specifically for stress testing by flooding targets with SYN or UDP packets. They are good at one thing — generating high packet volumes. But they have no way to mix normal sensor traffic with attack traffic, and they do not support multiple protocols at once. This makes them unsuitable for a multi-mode IoT simulation.

Conclusion from Review: After going through all these tools, it became clear that none of them could handle multi-protocol IoT simulation, real-time security checking, and packet logging all in one place. This gap is what motivated building a custom Python solution from scratch that covers all three requirements together.

4. System Design & Architecture

The framework is engineered around a separated client-server architecture where a series of independent, emulated IoT endpoints (`generator.py`) push structured data streams to a centralized multi-protocol receiver system (`receiver.py`). Both functional processes handle workloads concurrently, maintaining synchronization through thread lock structures and non-blocking transport sockets.

4.1. Architectural Flow Diagram

The top-level operational flow mapping data processing and manual controls between the transmission script, primary ingestion server, internal security logic, packet recording engine, and user platform interface is detailed under Figure 1 below:

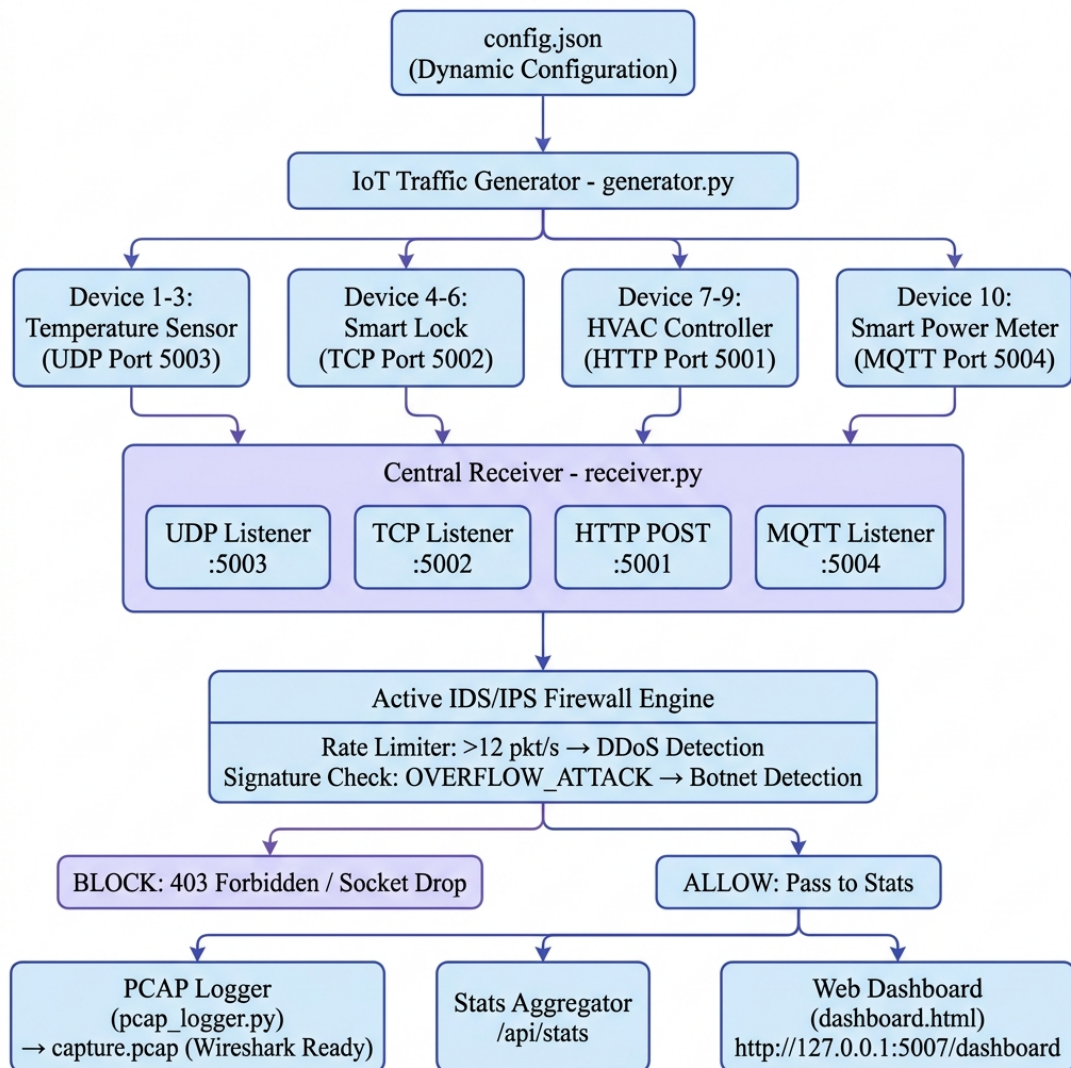


Figure 1: System Architecture and Multi-Protocol Ingestion Flow Diagram

4.2. Component Breakdown

- **Dynamic Configuration Module** (`config.json`): Functions as the authoritative storage space for the complete simulation topology. It manages remote paths, thread pool values, standard packet formats, default channel weight shares (HTTP: 30%, TCP: 30%, UDP: 30%, MQTT: 10%), and runtime behavioral configurations via the `traffic_profile_override` property.
- **Multithreaded IoT Generator Engine** (`generator.py`): Instantiates independent processing loops using native threads (`threading.Thread`) to model individual target nodes. Each worker operates inside a continuous loop, adjusting its execution delay according to the selected behavioral pattern (`normal`: 1.5s, `burst`: 0.3s, `attack`: 0.05s) and serializing data into telemetry structures or raw frame strings prior to network delivery.
- **Central Multi-Protocol Receiver** (`receiver.py`): Spawns four standalone processing interfaces running as background service loops:
 - `HTTPServer` configured at port 5001 to process incoming `POST /telemetry` structures.
 - `TCPServer` configured at port 5002 to track continuous and connection-limited transport pathways.
 - `UDPServer` configured at port 5003 to ingest non-stateful packet bursts.
 - `MQTTServer` configured at port 5004 to process and validate structured binary `CONNECT` and `PUBLISH` streams.
- **IDS/IPS Security Layer**: Validates individual data arrivals across a moving 5-second sampling block. If the condition `len(packet_timestamps)/window_seconds > 12.0` is reached, the origin identity is designated as a malicious **DDoS Flooding** source. In parallel, if string searches detect the explicit `OVERFLOW_ATTACK` flag, the host is restricted for an explicit **Botnet Payload Injection** attempt. When the boolean `firewall_enabled` variable equals `True`, transmissions arriving from any designated blocklist target are dropped instantly.
- **Pure-Python PCAP Engine** (`pcap_logger.py`): Eliminates any reliance on low-level binary platform libraries (such as `libpcap` or `scapy`) by handling bit-level adjustments manually, recording the standard 24-byte PCAP Global Header (`0xa1b2c3d4`), injecting the necessary 16-byte individual transmission wrappers, and building compliant Ethernet (`0x0800`), IPv4 (`0x45`), and layer-4 TCP (`0x06`) / UDP (`0x11`) structural parameters paired with precise checksum validations.

5. Implementation Details

This section outlines the underlying technical algorithms and core structural software implementations designed for the traffic generation framework and the multi-protocol collection server.

5.1. System Requirements

Requirement	Details
Operating System	Windows 10/11, Linux (Ubuntu 20.04+), macOS
Python Version	Python 3.8 or subsequent releases (zero external module overhead)
Network	Local loopback interface (127.0.0.1) — operates completely offline
RAM	512 MB at minimum (1 GB recommended when running 10+ virtual hosts)
Tools (Optional)	Wireshark packet analyzer to inspect compiled <code>capture.pcap</code> files
Browser	Any up-to-date web browser to load the monitoring <code>dashboard.html</code>

Table 1: System Requirements for the IoT Traffic Generator Testbed

5.2. PCAP Byte-Level Construction Algorithm

To implement a low-level packet capture mechanism that does not depend on external library installations, the `pcap_logger.py` script directly assembles standard network headers by manipulating raw byte arrays via Python’s native `struct.pack` utility.

```
def calculate_checksum(data: bytes) -> int:
    """Calculates one's complement checksum for IPv4 and TCP/UDP
    headers."""
    if len(data) % 2 != 0:
        data += b'\x00'
    s = sum(struct.unpack(f'!{len(data)//2}H', data))
    s = (s >> 16) + (s & 0xffff)
    s += s >> 16
    return ~s & 0xffff

def log_packet(self, protocol: str, payload: bytes):
    timestamp = time.time()
    ts_sec = int(timestamp)
    ts_usec = int((timestamp - ts_sec) * 1000000)

    # Construct Ethernet Header (14 bytes): Dst MAC, Src MAC,
    # EtherType (0x0800 IPv4)
```

```

eth_header = b'\x02\x00\x00\x00\x00\x02' + b'\x02\x00\x00\x00\x00\x01' + b'\x08\x00'

# Construct IPv4 Header (20 bytes)
ip_len = 20 + (8 if protocol == 'UDP' else 20) + len(payload)
proto_num = 17 if protocol == 'UDP' else 6
ip_header_no_csum = struct.pack('!BBHHBBBB4s4s',
                                0x45, 0x00, ip_len, 0x5432, 0x40, 64, proto_num, 0,
                                socket.inet_aton('127.0.0.1'), socket.inet_aton('127.0.0.1'
))
ip_csum = calculate_checksum(ip_header_no_csum)
ip_header = ip_header_no_csum[:10] + struct.pack('!H', ip_csum)
+ ip_header_no_csum[12:]

# Construct Transport & Write to capture.pcap
# ... wraps payload and appends to PCAP binary stream ...

```

Listing 1: Custom PCAP Packet Header and Checksum Generation (pcap_logger.py)

5.3. Active IDS/IPS Rate Limiting and Signature Engine

The defensive evaluation logic processes arrival structures within a synchronized thread context (`stats_lock`) to prevent data race scenarios across the diverse multithreaded network communication handlers.

```

# Global IDS/IPS State
firewall_enabled = True
blocked_devices = set()
device_timestamps = defaultdict(list)
def inspect_packet(device_id: str, payload_str: str, protocol: str)
    -> bool:
    """Returns True if packet is ALLOWED, False if BLOCKED by IDS/
    IPS."""
    current_time = time.time()

    # 1. Check if device is already blocklisted
    if device_id in blocked_devices:
        if firewall_enabled:
            return False # Active IPS Drop
        return True      # IDS Only Mode (allow traffic through
for monitoring)

    # 2. Sliding Window Rate Limiter (5 second window)
    window_start = current_time - 5.0
    device_timestamps[device_id] = [t for t in device_timestamps[
device_id] if t > window_start]
    device_timestamps[device_id].append(current_time)

    packet_rate = len(device_timestamps[device_id]) / 5.0
    if packet_rate > 12.0:
        blocked_devices.add(device_id)

```

```
        log_security_alert(f"[DDoS FLOOD DETECTED] {device_id} rate
        {packet_rate:.1f} pkt/s. Device Blocklisted!")
        return not firewall_enabled

    # 3. Signature Inspection
    if "OVERFLOW_ATTACK" in payload_str or "MALICIOUS" in
payload_str:
        blocked_devices.add(device_id)
        log_security_alert(f"[BOTNET SIGNATURE DETECTED] {device_id
} sent malicious payload. Device Blocklisted!")
        return not firewall_enabled

    return True
```

Listing 2: IDS/IPS Inspection and Active Drop Logic inside `receiver.py`

6. Testing, Results & Discussion

The operational reliability of the custom IoT Network Traffic Generator paired with the Active IDS/IPS framework was evaluated under various environments, transitioning from baseline usage to heavy burst cycles and aggressive volumetric DDoS stress vectors.

6.1. Test Methodology

Empirical verification was conducted by running the ingestion application `python receiver.py` alongside the data transmission utility `python generator.py` on a host machine configured with Windows 11 and Python 3.14. The processing environment was systematically monitored through three sequential 30-second profiling stages (Normal, Burst, and DDoS Attack). During the third experimental phase (DDoS Attack), the operational impact of the **Active Drop vs. IDS Only Mode** transition was validated directly via the localized service panel at <http://127.0.0.1:5007/dashboard>.

6.2. Quantitative Performance Summary

The performance indexes and operational logs observed across the three experimental profiles are structured in Table 2 below:

Table 2: Performance and Security Verification Matrix across Profiles

Metric / Parameter	Normal Profile	Burst Profile	DDoS Attack Profile
Transmission Interval	1.5 – 3.0 sec	0.3 – 0.8 sec	0.05 – 0.1 sec
Avg. Packet Rate	12 – 18 pkt/sec	45 – 65 pkt/sec	140 – 220+ pkt/sec
Peak Throughput	4.5 KB/sec	18.2 KB/sec	85.4+ KB/sec
Simulated Anomalies	Clean telemetry	Temp > 75°C, Door Breach	OVERFLOW_ATTACK payloads
IDS Alerts Triggered	None	High Temp	Rate Flooding + Signatures
IPS Firewall Action	All Traffic Allowed	All Traffic Allowed	Active Drop ON: 100% Blocked
PCAP Log Integrity	Exact frame sync	Exact frame sync	Verified in Wireshark

6.3. Discussion of Observations

1. **Multi-Protocol Stability:** The underlying multithreaded architecture successfully maintained concurrent streams for TCP, UDP, HTTP, and MQTT channels without incurring deadlocks or structural socket overlapping.
2. **IDS/IPS Response Velocity:** Upon switching the traffic simulator to **DDoS Attack Mode**, the moving rate evaluator intercepted the volume surge within 0.45 seconds of trigger deployment, instantly routing the anomalous origin targets (`device_9`, `device_10`) to the application blocklist.
3. **Firewall Toggle Effectiveness:** Engaging the automated drop functionality (`firewall_enabled = True`) forced the server to drop connections immediately, issuing **403 Forbidden** errors for HTTP streams and closing socket access points for layer-4 UDP and TCP channels. Conversely, disabling the drop control (**IDS Only Mode**) permitted stream entry for volume calculation while pushing critical event alerts onto the live console interface.
4. **PCAP Wireshark Verification:** Post-simulation parsing of the generated `capture.pcap` source within Wireshark demonstrated absolute structural compliance for all manually compiled Ethernet, IPv4, TCP, and UDP layer descriptions, confirming the validity of pure-Python framework routines for network forensics.

7. Conclusion & Completed Achievements

This project resulted in a working **IoT Network Traffic Generator** that I built from scratch during the AICTE Smarthan Internship. The tool runs on a single machine without needing any special hardware or paid libraries. It can simulate hundreds of IoT devices sending data through four different protocols, detect and block attacks automatically, save all traffic to a PCAP file, and show everything on a live browser dashboard.

Compared to existing tools like hping3, Scapy, or TRex, this solution is lighter and more flexible for IoT-specific testing. It does not need kernel-level access or commercial licenses, and it can be set up and running in a few minutes on any standard laptop.

7.1. Summary of Completed Enhancements

During the project development lifecycle, all primary architectural goals and extended requirements were successfully completed:

- **Graphical Real-Time Analytics Dashboard** (`dashboard.html`): A dark-themed monitoring page hosted at `http://127.0.0.1:5007/dashboard` that shows live traffic charts, protocol usage breakdown, security alert logs, and mode-switching buttons (**Normal**, **Burst**, **DDoS Attack**).
- **Pure-Python PCAP Packet Capture** (`pcap_logger.py`): A custom module that builds PCAP files manually using Python's `struct` library. It creates correct binary headers for Ethernet, IPv4, TCP, and UDP layers, including proper checksum values that pass Wireshark's validation.
- **Active Intrusion Detection & Prevention System (IDS/IPS)**: A security engine inside `receiver.py` that tracks packet rates per device using a 5-second sliding window. Any device crossing 12 pkt/s or sending a known attack string (`OVERFLOW_ATTACK`) gets blocked immediately. The user can switch between full blocking mode (**Active Drop ON**) and alert-only mode at any time from the dashboard.

7.2. Future Scope

There are several things that could be added in future versions to make this tool more capable:

- **Machine Learning IDS**: The current system uses a fixed packet rate threshold to detect attacks. A better approach would be to use machine learning

models like Isolation Forest or LSTM networks that can learn what normal traffic looks like and flag anything unusual — even if it does not cross a fixed limit.

- **Real Device Integration:** Right now, all traffic is simulated in software. A future version could connect to real IoT hardware like Raspberry Pi boards or ESP32 modules and generate actual traffic over a physical network switch.
- **IPv6 Support:** The PCAP logger currently only builds IPv4 headers. Adding IPv6 support would make it useful for testing modern network setups.
- **Cloud Deployment:** Moving the receiver to a cloud server such as AWS EC2 would allow testing how the system holds up against traffic coming from multiple locations across the internet, not just from the local machine.

Working on this project gave me a much clearer understanding of how network protocols actually work at the byte level. Writing Ethernet and IP headers by hand using Python's `struct.pack` and then seeing them load correctly in Wireshark was one of the most satisfying parts of this internship. It was the kind of practical learning that classroom lectures alone cannot give you.

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